Rank-27 Vivify and Sweep Lifecycle Repair

Tactix engineering handoff

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Executive outcome

The suspicious variable-range Vivify policy was real and has been removed. Vivify now follows the retained Kissat evidence for candidate scope, cumulative tier allowance, jumped-reason deduction, terminal effects, productivity, and its interleaving with the surrounding Probe pipeline. The first later portable divergences were then repaired in sequence: TransitiveReduce source order and Sweep's semantic-dose/representation lifecycle.

The Sweep defect was a category error rather than another benchmark gate. Tactix charged a whole-arena representation rebuild to the current live Sweep allowance, while Kissat's source limits that call only by embedded `kitten_ticks`. The rebuild ended a productive retained tail after the first equivalence batch. Tactix now uses Kitten ticks for the live semantic ceiling, continues to price representation work in later scheduler accounting, and carries one dense substitution ledger across productive environments before collecting the arena once at call exit.

Release passes 72/72. The required matched 28B+5C guard passes 66/66 with all models and proofs checked and zero certificate residue. The first meter-only candidate had timed out on the fixed Tier-C `T105.2.0` guard; the complete lifecycle repair proves it UNSAT in 1,043.578 seconds. Immutable rank 27 still returns UNKNOWN at 1,200.004 seconds, so rank 27 remains open.

What was wrong

At the common conflict-3069 state, retained Kissat evidence continues the second Sweep call through 2,303 resumed seeds and derives 343 equivalences. Tactix's original live meter was

$$\texttt{sweep_ticks_sub} + \texttt{sweep_ticks_repr}.$$

After the first productive environment, Tactix crossed its variable-width arena bridge. That one collection charged roughly 18.8 million representation words against a 19.19-million semantic dose and stopped the call. Kissat has no corresponding charge: `sweep.c` sets and tests the outer allowance in cumulative `kitten_ticks` alone.

Separating the live meter restored the semantic tail exactly, but initially exposed a second host-representation cost: Tactix collected the entire arena after every productive environment. The resulting call charged 1,100,956,356 representation words, took 53.072 seconds, and failed the full guard. This failed guard correctly blocked a Tier-D witness.

The existing `SweepSubstitutionSession` supplied the general repair. Its ordered dense occurrence ledger is updated by every substitution transaction, so the next environment already sees the previous environment's consequences. The dense call therefore needs one collection at exit, not one collection per productive environment. Sessionless/public calls retain immediate collection because they own raw arena references and must refresh them after relocation.

Implementation and focused evidence

The source change has three parts.

- `sweep_semantic_effort` is the cumulative Kitten counter used by the current call's ceiling and remaining per-environment allowance.
- `sweep_effort` remains Kitten ticks plus representation words for policy scheduling, so real host work is not hidden.
- A connected dense call collects once after its ordered environment sequence; a sessionless call keeps its immediate collection and generation refresh.

No benchmark name, hash, rank, literal, deadline, variable range, or tuned constant enters solver behavior. A focused two-environment test requires at least two equivalences and substitutions but exactly one arena generation increment under a connected dense index. The existing sessionless test pins the opposite correctness-required lifecycle.

Candidate-only job `1637687` gives the following bounded comparison.

Measure	per-environment collection	one call collection
Equivalences	343	343
Emitted binaries	686	686
Retired variables	369	369
Representation words	1,100,956,356	4,720,714
Sweep time	53.072 s	24.570 s
Final active / unassigned	429,499 / 429,303	429,499 / 429,303

The optimized call visits 2,306 retained seeds rather than 2,303. The extra three are zero-effect: the equivalence set, binary count, retirements, and final slot state are unchanged. This count is stated as the next scheduling detail, not hidden or claimed exact.

Full correctness guard

Matched campaign `benchmarks/runs/20260821-2145-BC-sweep-deferred-bridge/`, job `1637688`, ran the accepted pin and candidate on all 28 Tier-B fixtures plus the fixed five mechanism-relevant Tier-C fixtures under 16 GiB and 1,200 seconds.

Arm	valid	SAT / UNSAT	aggregate	peak RSS
Accepted pin	33/33	17 / 16	2205.151 s	3.234 GiB
Candidate	33/33	17 / 16	2924.991 s	3.081 GiB

All 34 SAT models and 32 UNSAT proofs verify. There are no invalid, missing, malformed, cleanup-bad, or residual-certificate tasks. The candidate ratio is 1.326436, so this is not promoted as a new accepted performance pin. It does, however, remove the solve loss that rejected the first candidate: `T105.2.0` is proof-verified UNSAT in 1,043.578 seconds, compared with the accepted pin's 790.936 seconds. Both proofs were deleted after checking.

Immutable rank-27 witness

Only after the complete guard passed was job `1637757` submitted. It returned UNKNOWN at 1,200.004 seconds. The runner recorded a 196,083,712-byte partial proof, SHA-256 `b5383d30...54a15`, then deleted it and all checker scratch. Certificate residue is zero. Rank 27 is not added to the nine closed ranks.

Experiment discipline and next action

The immutable input SHA-256 is `5435b1b2...bb4e9`; the accepted binary is `b6c991e0...fb6f`; the candidate is `f9206a95...d82bb9`. The branch and `origin/main` were synchronized before work, surviving worktrees and unrelated jobs were left untouched, and no frozen reference solver was rerun. Diagnostics used the retained Kissat trace and candidate-only Tactix jobs.

The next rank-27 localization retains the closed Vivify, TransitiveReduce, and Sweep consequence/transaction repairs. It begins with the three zero-effect retained seeds, then follows the earliest downstream common-state divergence. Any further source change again owes the complete 28B+fixed-C guard before a Tier-D witness.

The durable technical report is `benchmarks/reports/2026-08-21-tierd-rank27-sweep-reference-lifecycl md`. The current resume surface is `PROGRESS.md`; mechanism decisions are in `docs/implementation-notes.md` and `docs/scheduling-parity-ledger.md`.